

ECEN 607: Advanced Analog Circuit Tech

ASSN 2:REPORT

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ECEN-607: Advanced Analog IC Design

Assignment #2

Design at transistor level a two-stage amplifier; use the following specifications:

VDD = 1.8 V

VSS = 0 V

DC Gain > 40 dB

Employ miller compensation to stabilize the amplifier; loop phase margin > 45°

. Your amplifier

must also satisfy the following specifications:

GBW > 4 MHz

Load Capacitor = 100-500 pF (amplifier's phase margin must be maintained over 45 degrees for the entire range of CL).

VDSAT for the transistors, should be in the range {150mV – 250mV}.

Current sources needed for biasing the amplifier stages can be implemented employing ideal current sources.

Results:

i) Show amplifier's functionality. Simulate the amplifier and include in your report screen shots of both magnitude and phase response. Report the values of the unity gain frequency, phase margin and gain margin.

ii) Implement a capacitive amplifier with an inverting gain of 6dB, $\beta=1/3$; minimum capacitor used in the capacitive amplifier must be $\geq 100\text{fF}$. You may need to connect a large resistor ($>1\text{Mohm}$) in parallel with the feedback capacitor; why?

a. Simulate the amplifier and report both magnitude and phase response.

b. Plot the step response for an input step of 0 to 100mV variation. Report the 1% settling time and value of the settling error. Ideal output voltage variation should be -200mV.

c. Comment on the results.

iii) Repeat ii) for the case of an inverting amplifier with gain=12dB; evidently the feedback factor is different now. For the transient simulation, reduce the size of the input step to 0-50mV.

Please discuss your results; include necessary screen shots.

Required tables:

Sizing of transistors:

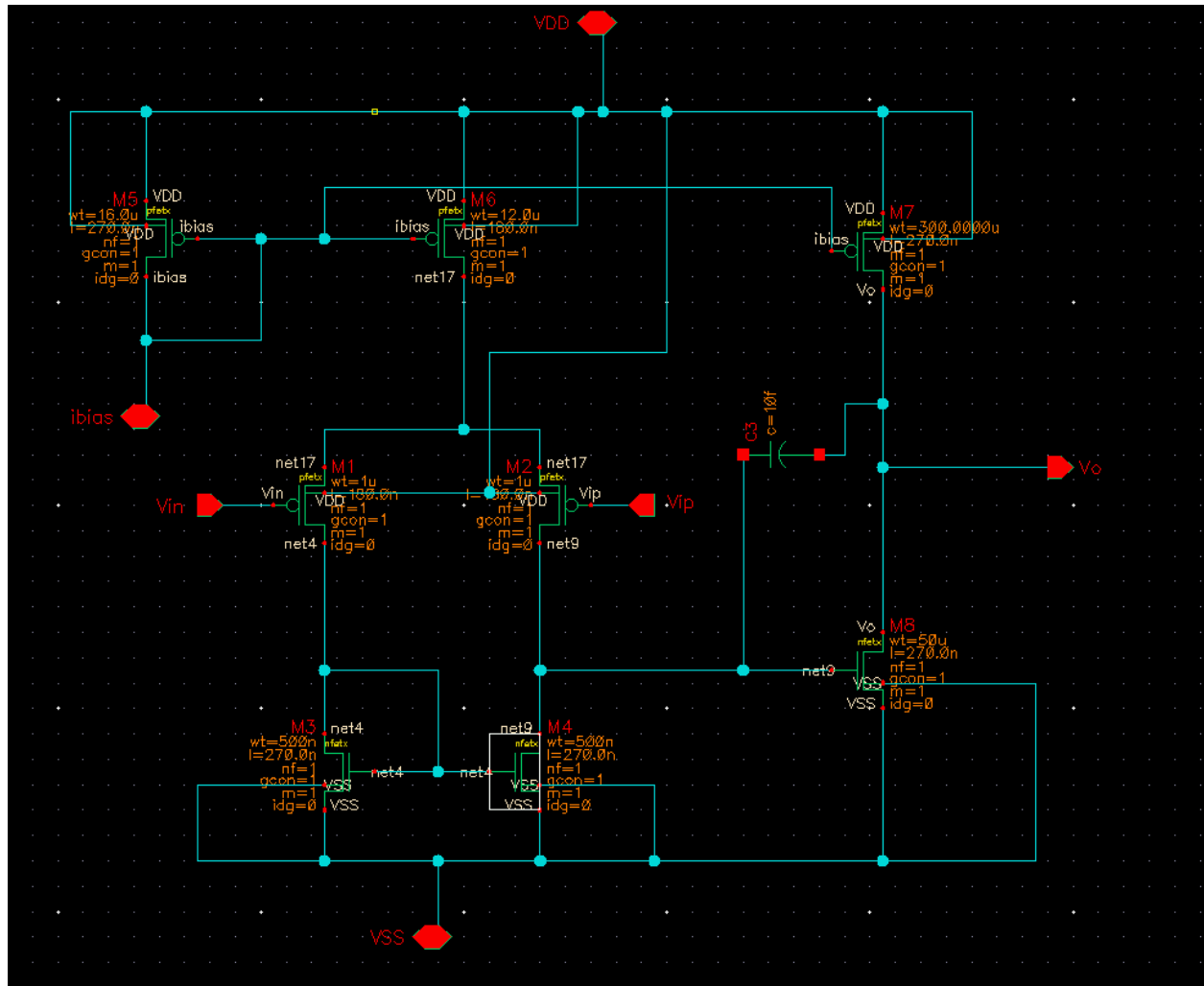
M1	1u	180n
M2	1u	180n
M3	500n	270n
M4	500n	270n
M5	16u	270
M6	12u	180
M7	300u	300u
M8	50u	50u

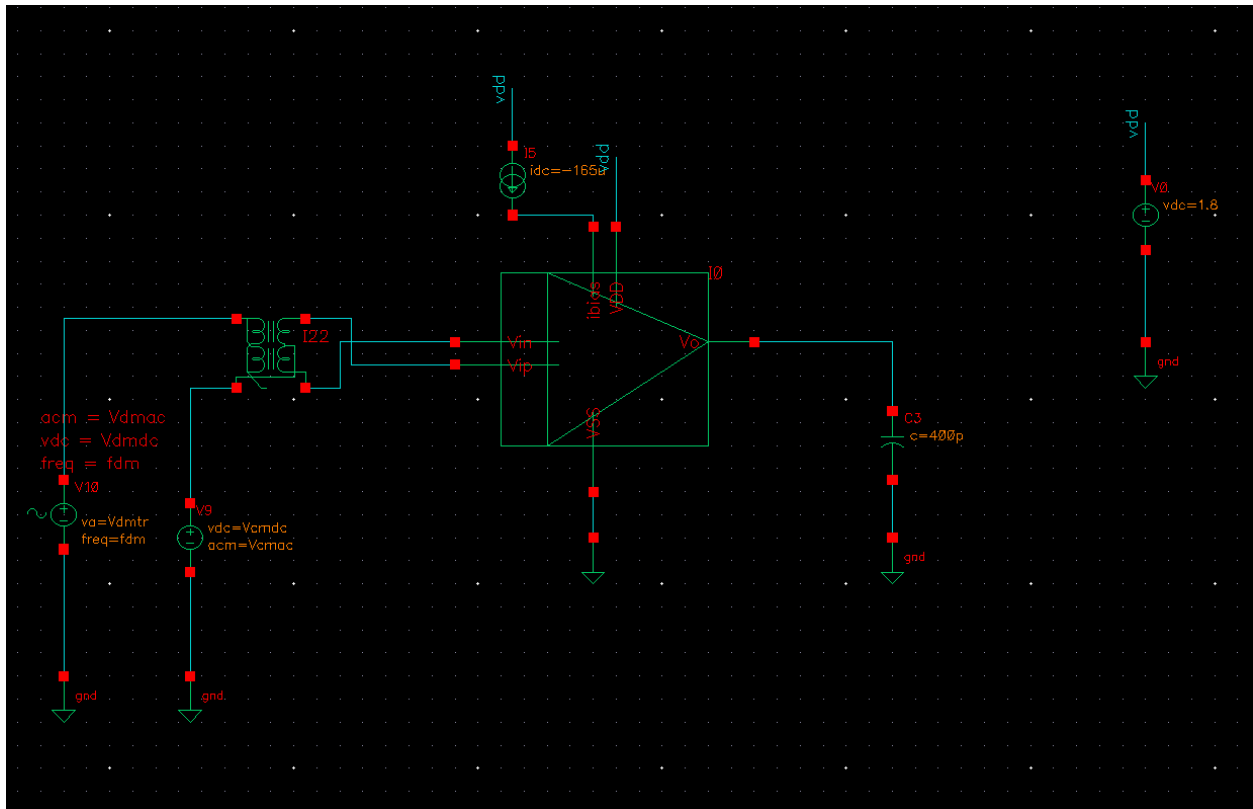
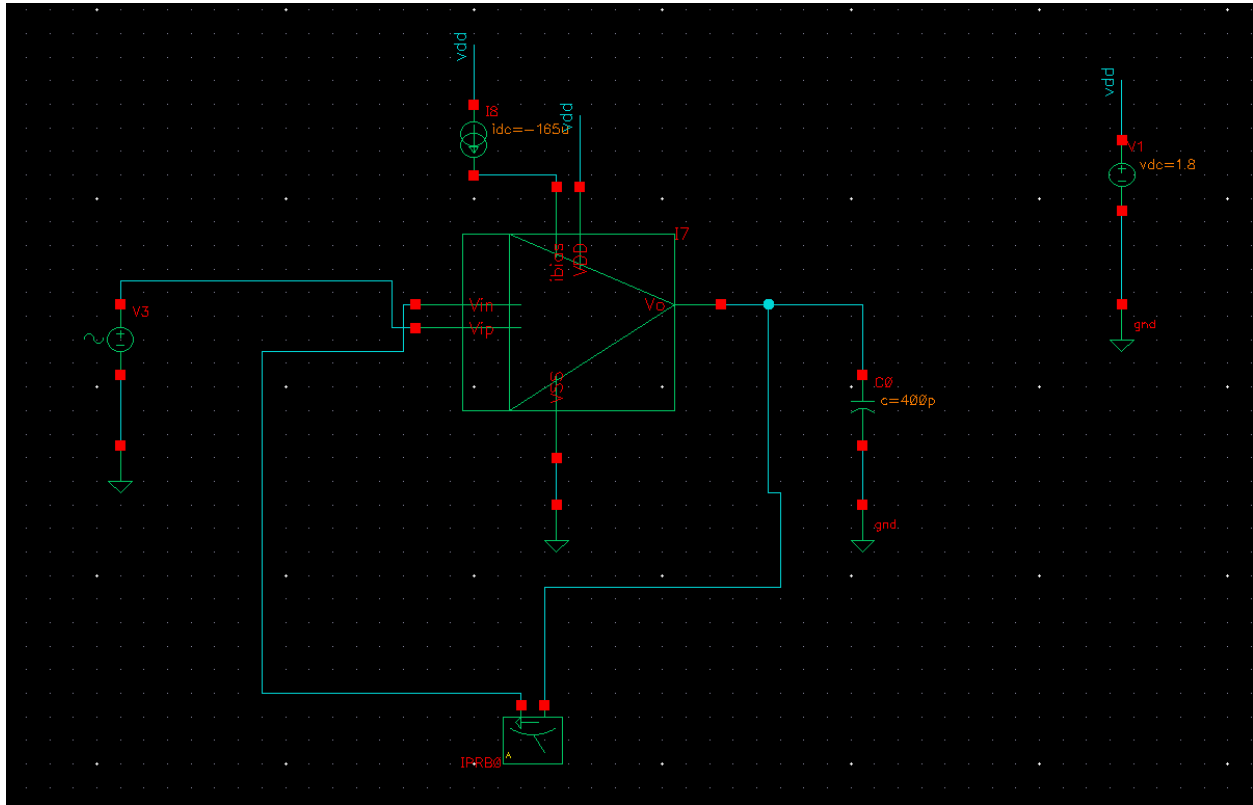
Obtained Specifications:

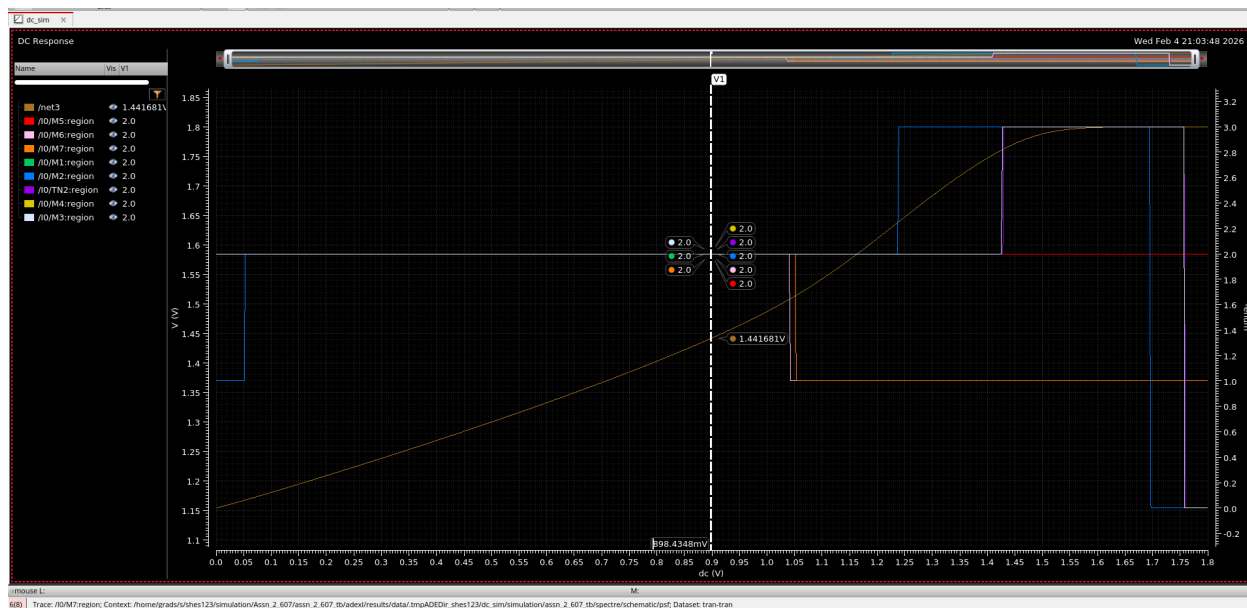
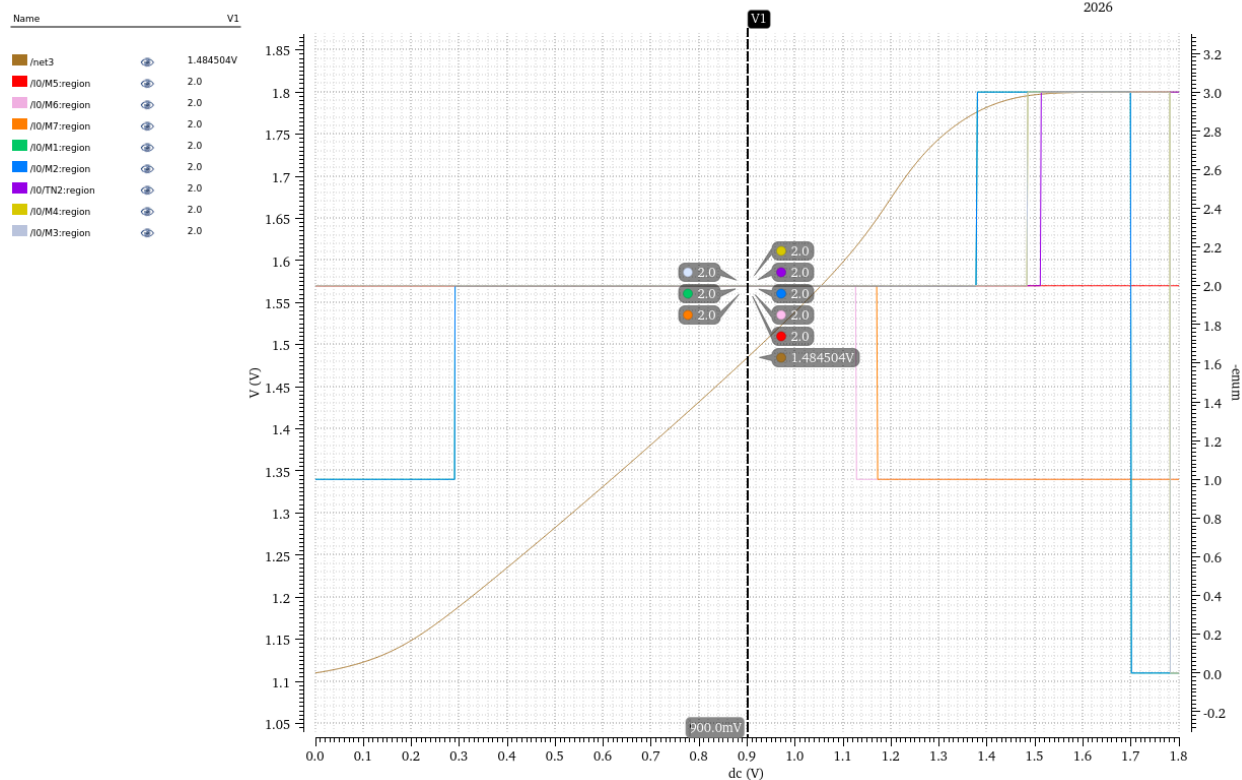
Gain	42.116db
GBW	29.17MHz
ID	165 u
GAIN MARGIN	-125.56 db
PHASE MARGIN	143.5 deg
CC	10pF
CF -6DB	100fF
CIN-6DB	100fF
R	2M ohm
CF -12DB	100fF
CIN-12DB	200fF

i) Show amplifier's functionality. Simulate the amplifier and include in your report screen shots of both magnitude and phase response. Report the values of the unity gain frequency, phase margin and gain margin.:

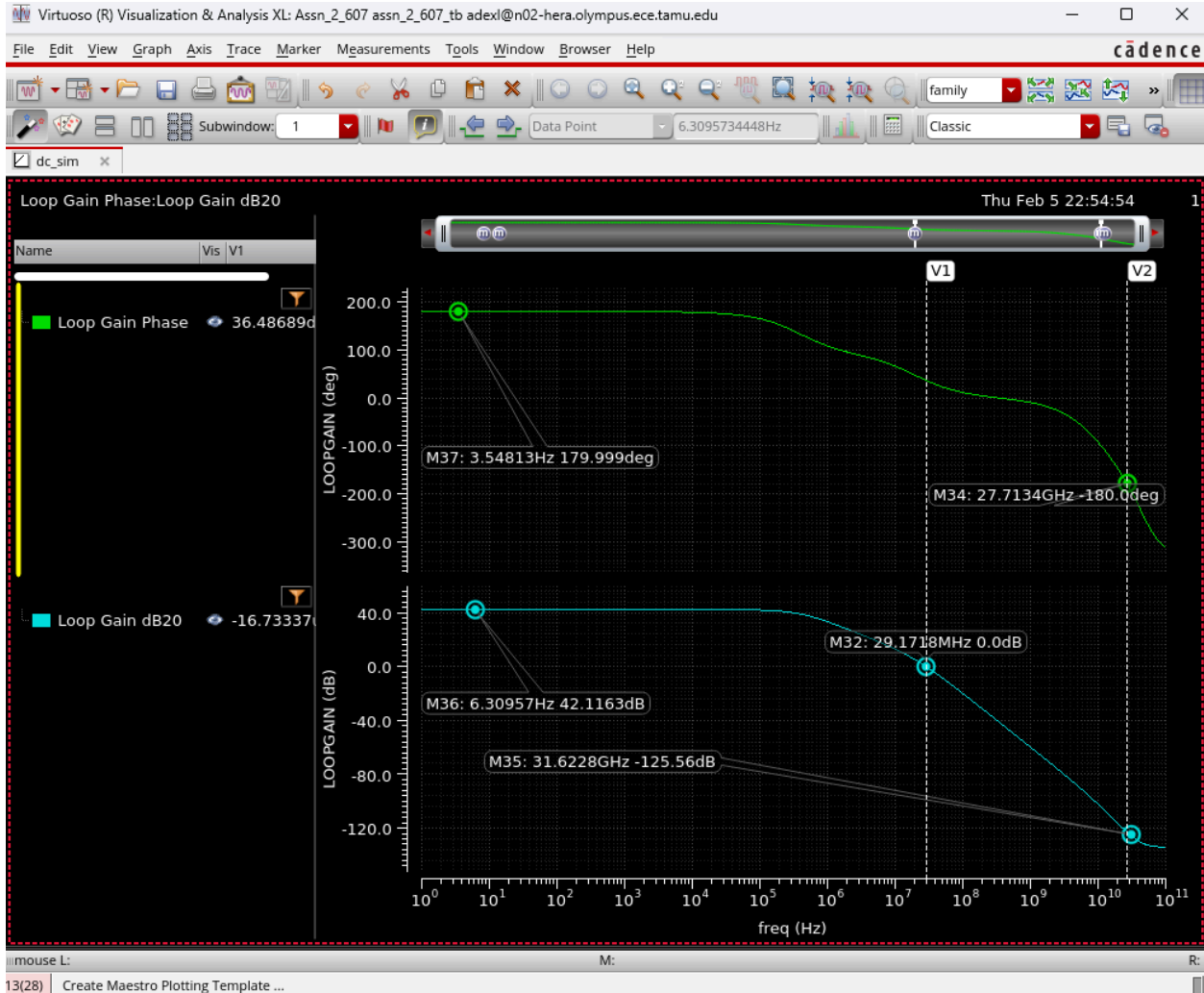
Schematics:







Gain and phase using iprobe:



ADE XL Test Editor - dc_sim@n02-hera.olympus.ece.tamu.edu

Session Setup Analyses Variables Outputs Simulation Results Tools Help

cadence

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Design Variables

Name	Value
1 Vdmdc	0
2 Vdmac	0
3 Vdmtr	0
4 fdm	0
5 Vcmdc	900m
6 Vcmac	1
7 p5vonly	0

Analyses

Type	Enable	Arguments
1 dc	☐	t 0 1.8 1m Linear Step Size Start-Stop /V9
2 ac	☐	1 100G 20 Logarithmic Points Per Decade Start-Stop
3 stb	☒	1 100G 20 /IPRB0 Logarithmic Points Per Decade Sta...

Outputs

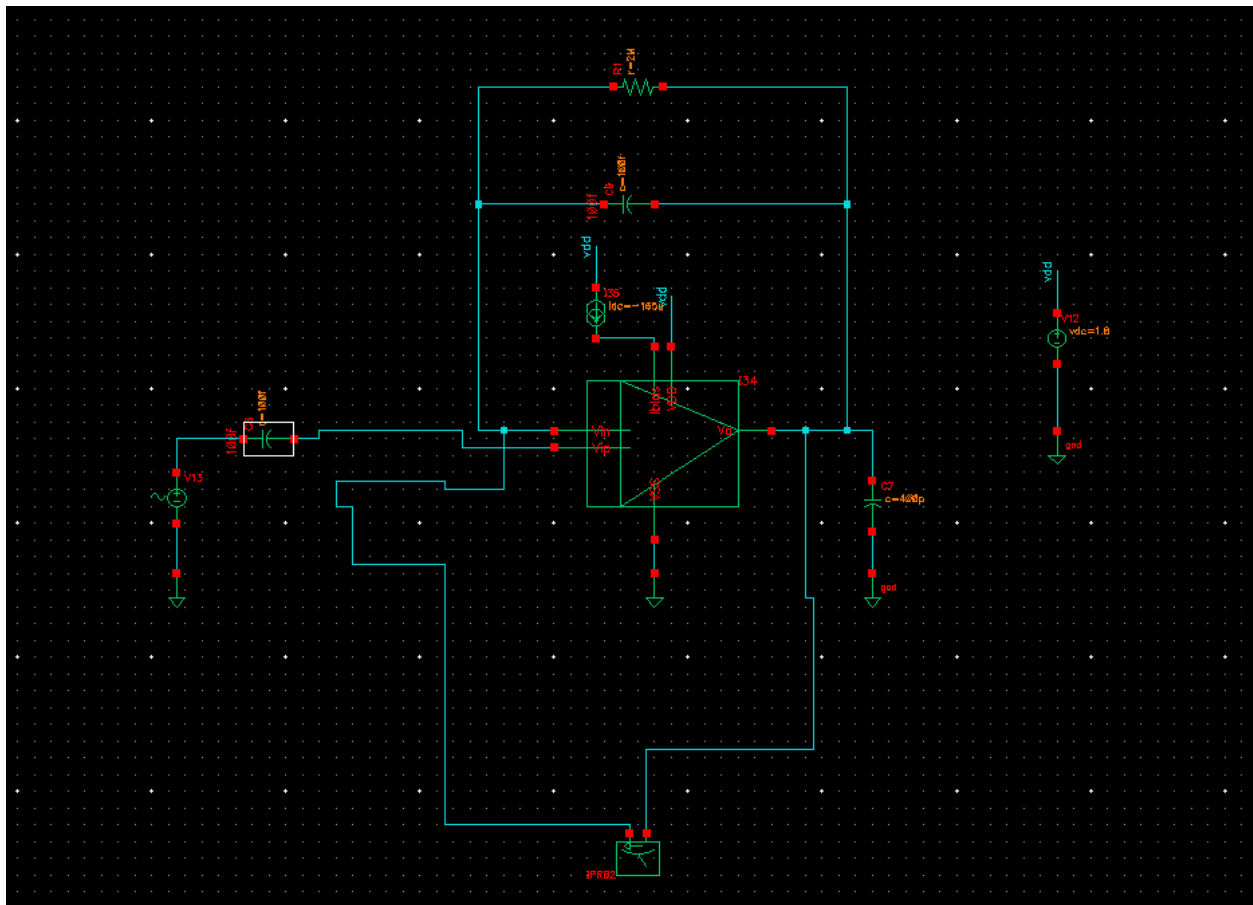
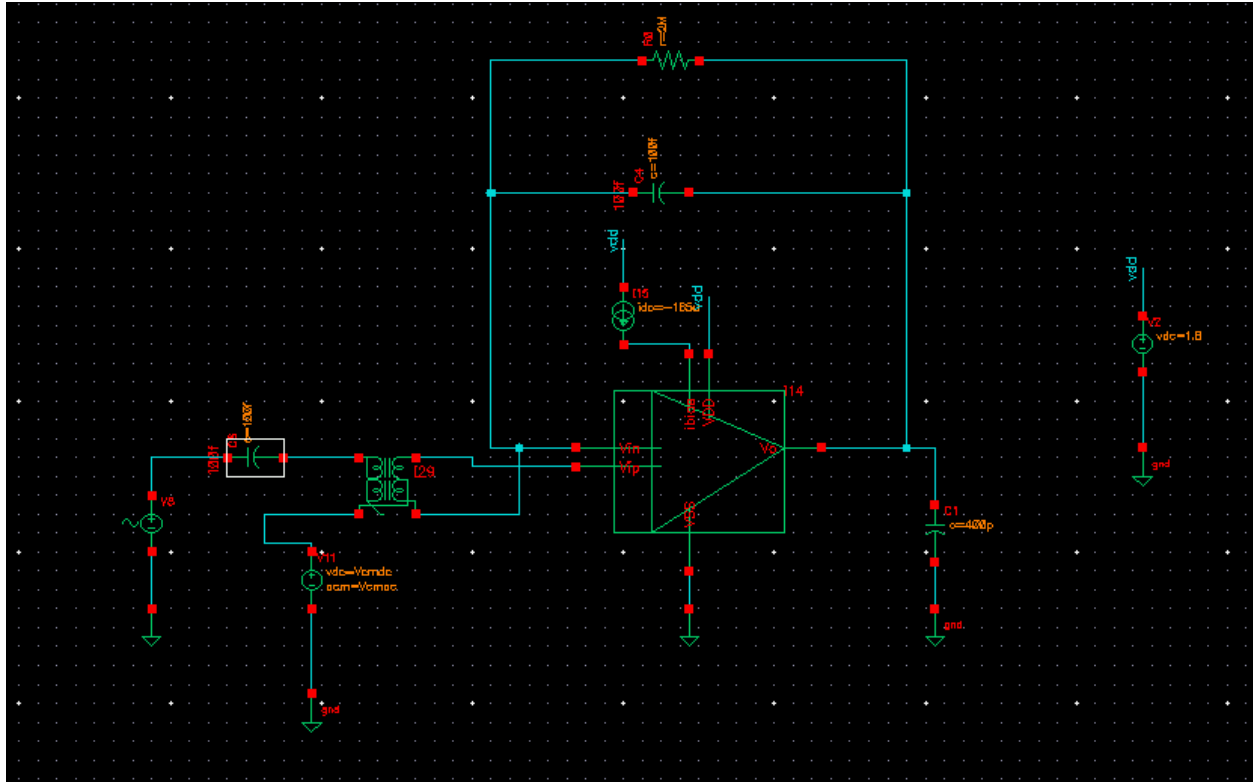
Name/Signal/Expr	Value	Plot	Save	Save Options
4 phase margin		☐	☐	
5 gain-margin		☐	☐	
6 Loop Gain Phase	wave	☒	☐	
7 Loop Gain dB20	wave	☒	☐	
8 net5		☐	☒	allv
9 gain_margin_1	-123	☒	☐	
10 phase_margin_1	143.5	☒	☐	

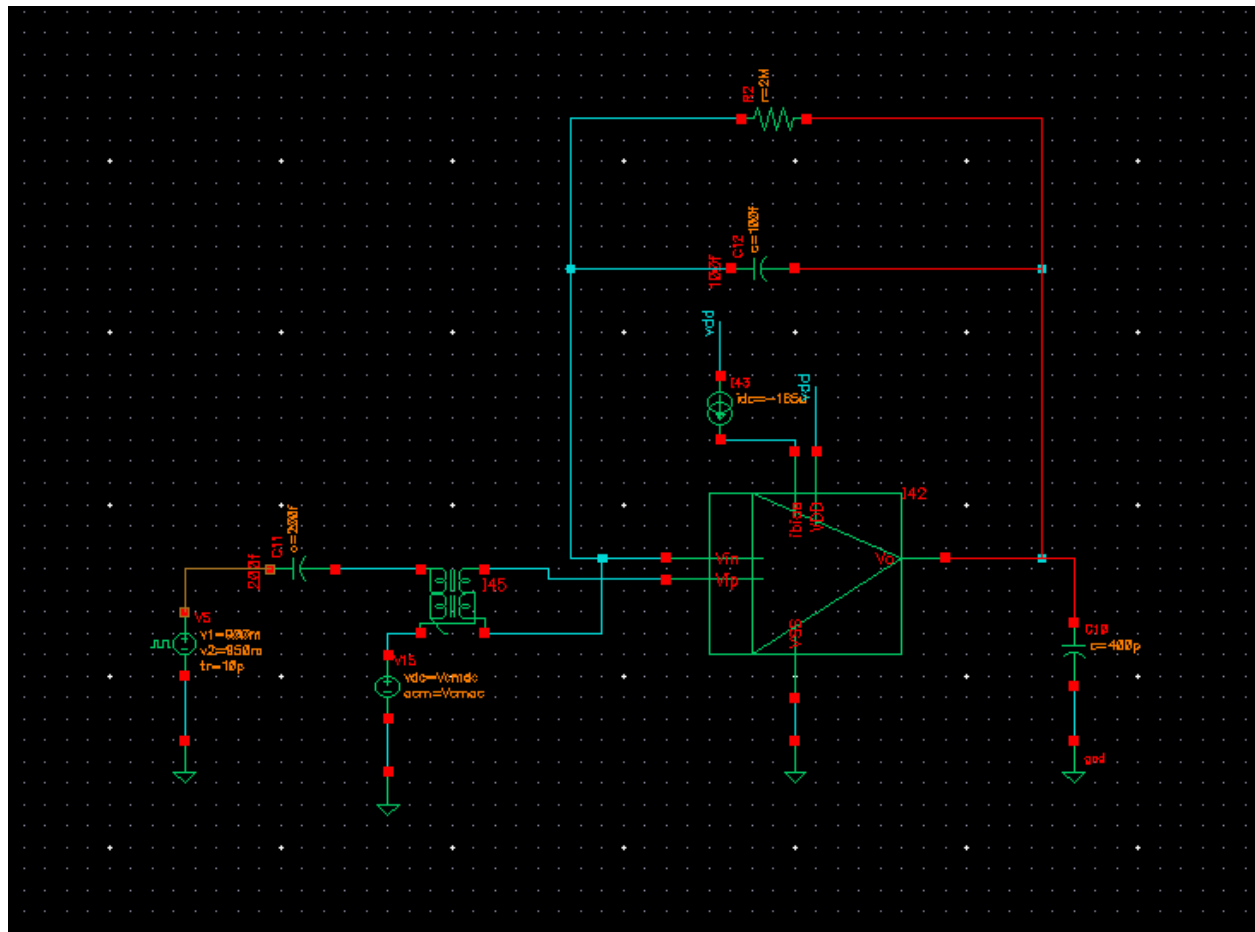
> Results in ...shes123/dc_sim/simulation/assn_2

12(25) | Model Libraries ... | Status: Ready | T=27 C | Simulator: spectre aps | State: dc_sim_active

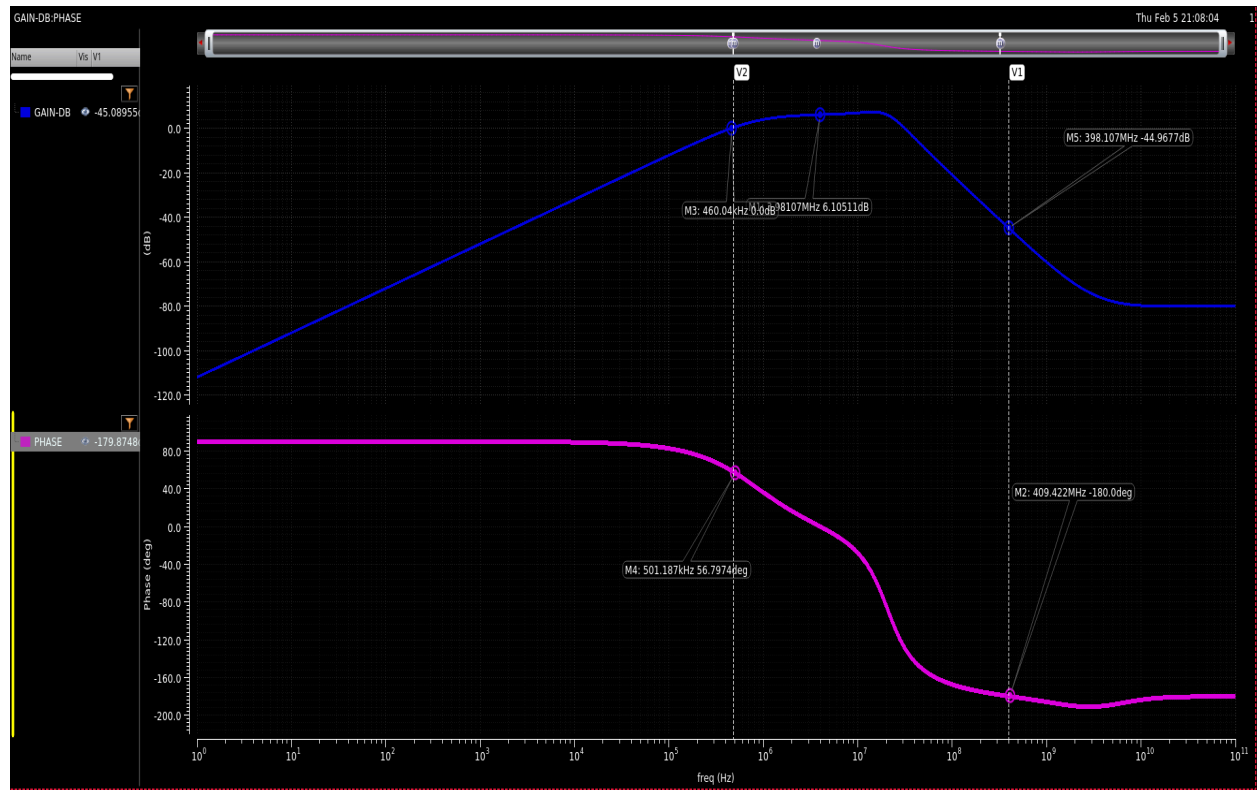
ii) Implement a capacitive amplifier with an inverting gain of 6dB, $\beta=1/3$; minimum capacitor used in the capacitive amplifier must be $\geq 100\text{fF}$. You may need to connect a large resistor ($>1\text{Mohm}$) in parallel with the feedback capacitor; why?

Schematic setups for 6db gain:

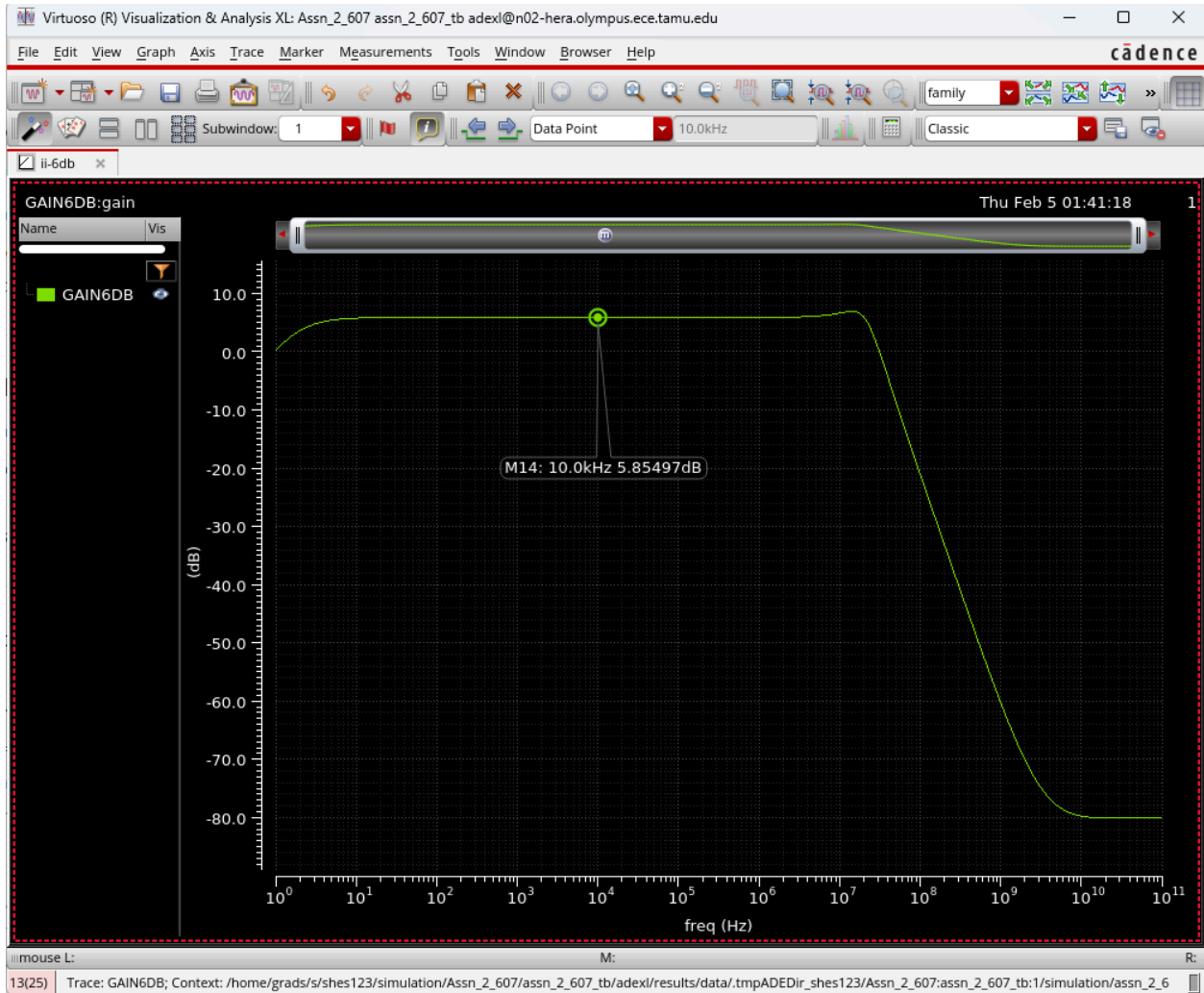




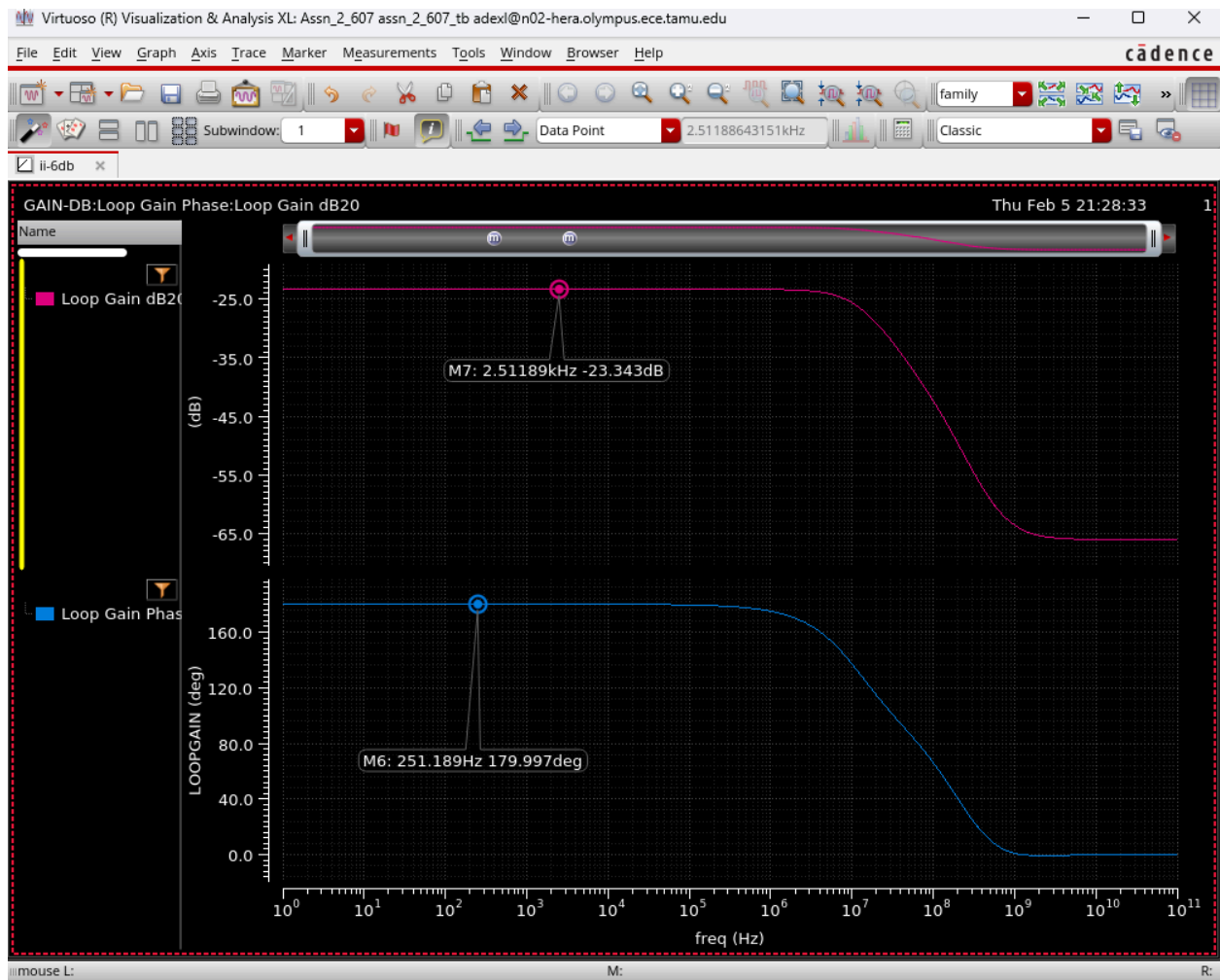
WHEN $R=2M$ OHM:



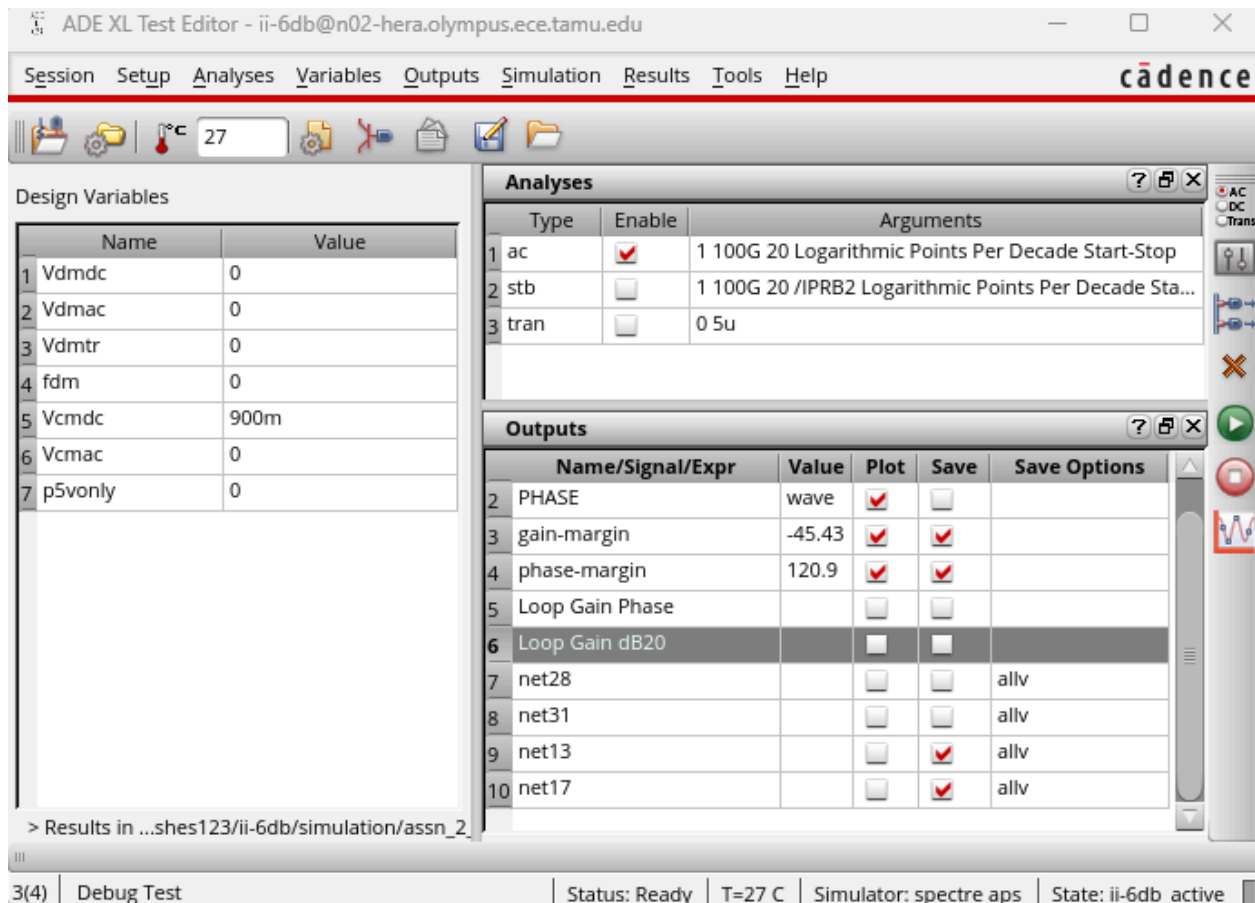
When R is very large:



Using iprobe:
THE LOOP GAIN AND PHASE:



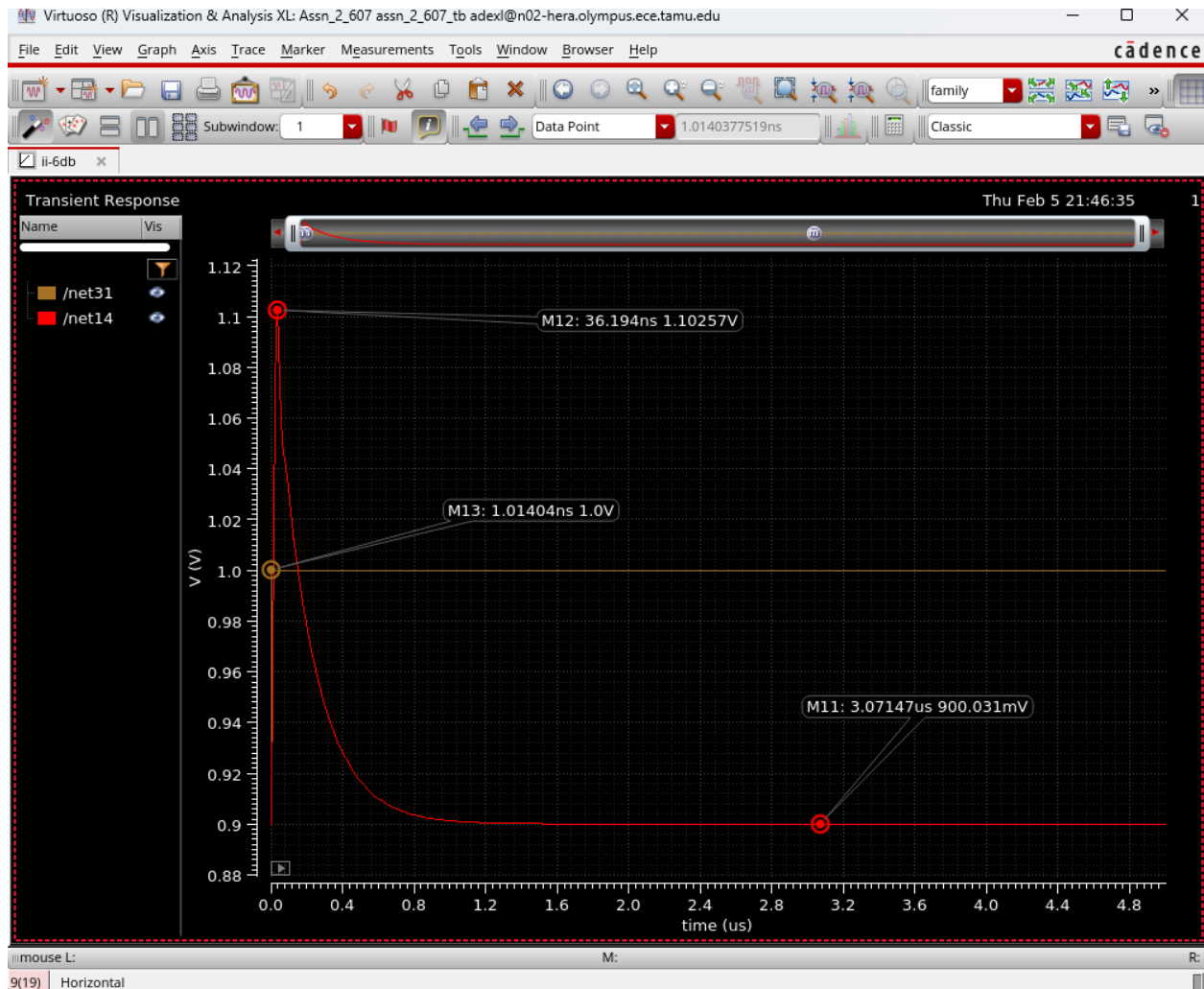
Phase margin and gain margin for the 6db gain:



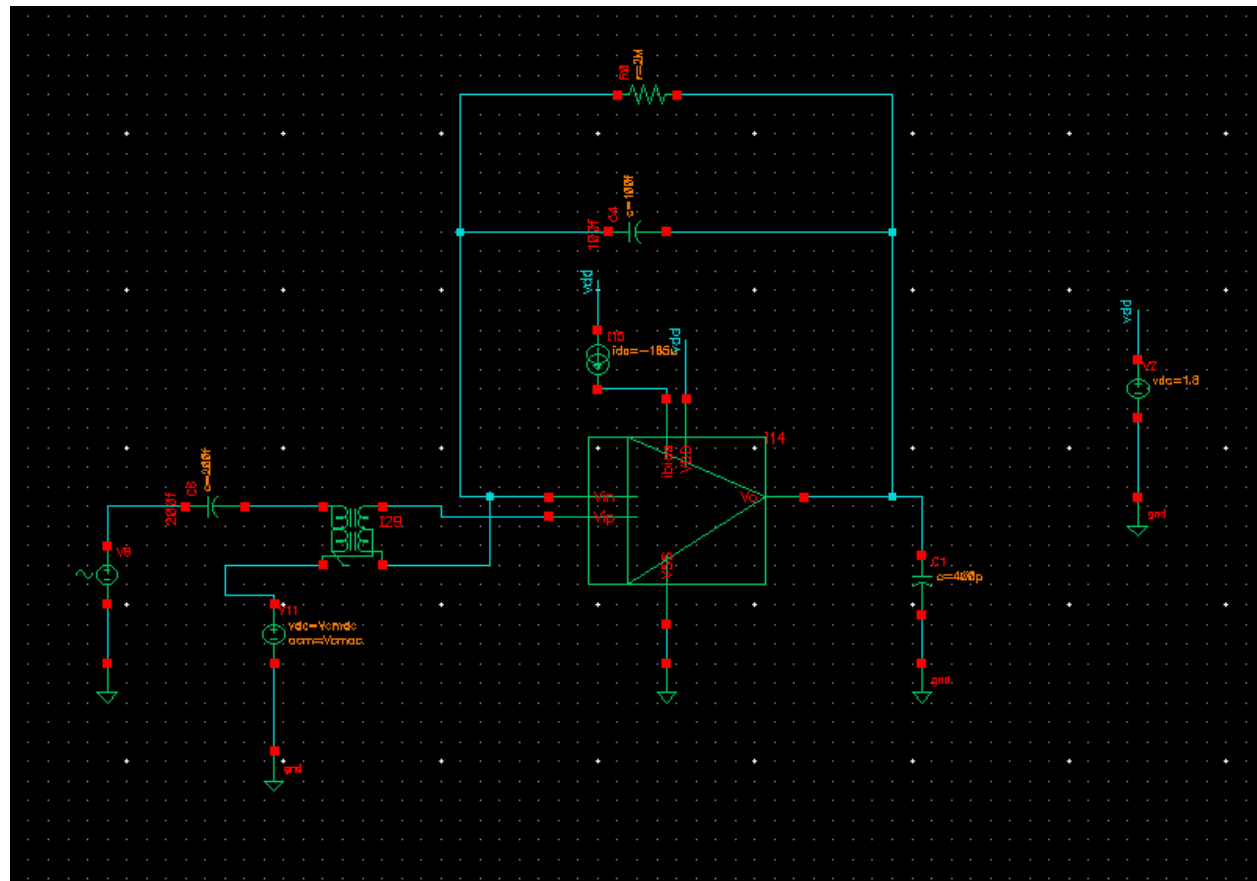
Step response with top 1 percent settling time

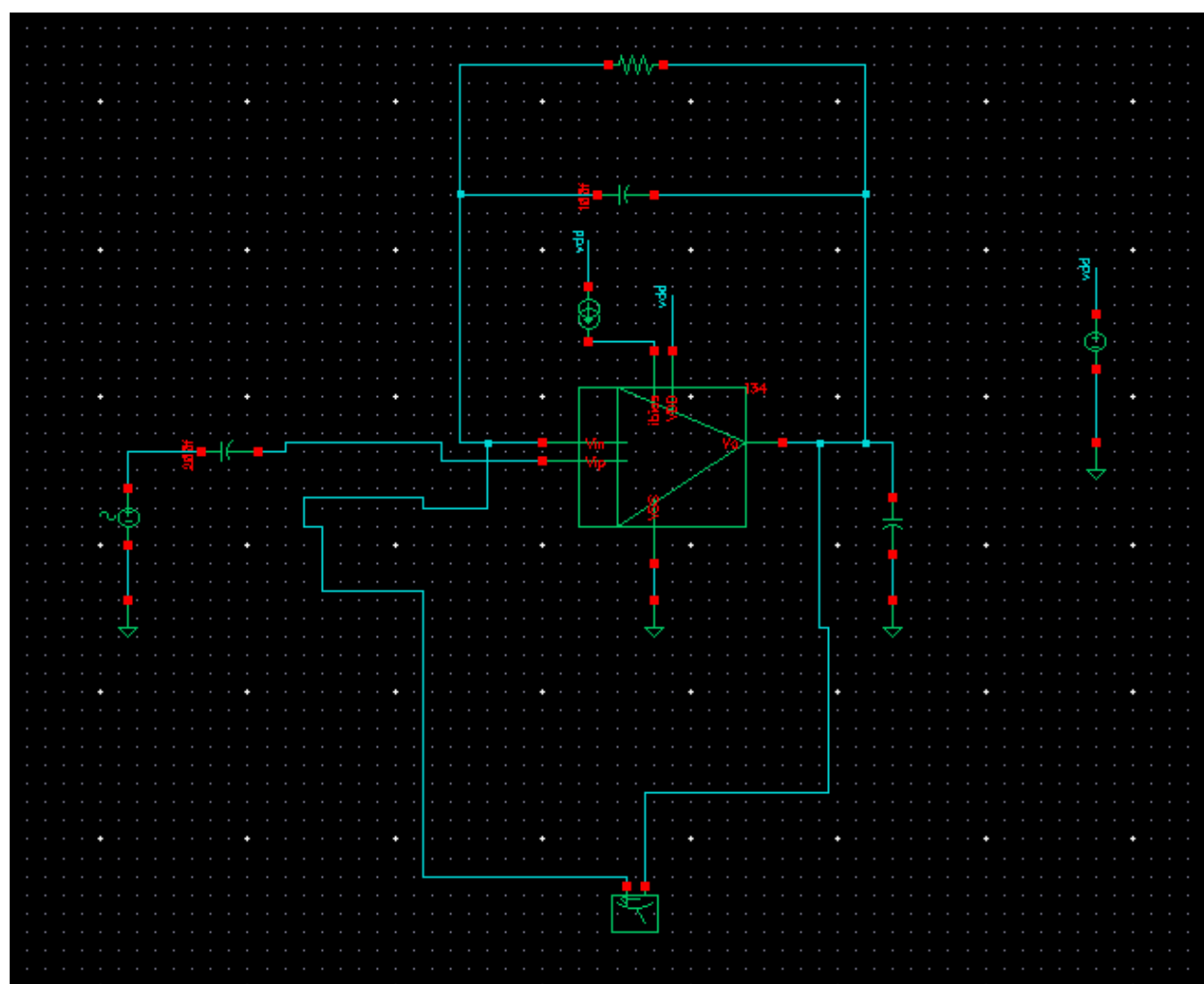
For a 0 to 100 mV step input, the ideal output change is -200 mV (from 1.09 V to 0.89 V). The simulated step response shows the output rapidly drops from 1.09 V and settles at approximately 0.9 V within about 1 μ s, corresponding to a voltage change of 190 mV. The output does not enter the 1% settling band (0.888 V – 0.892 V), and thus the 1% settling time is not achieved. The steady-state error is approximately 10 mV above the ideal value. This indicates a small, but non-zero, DC error in the output (likely due to finite open-loop gain or load effects).

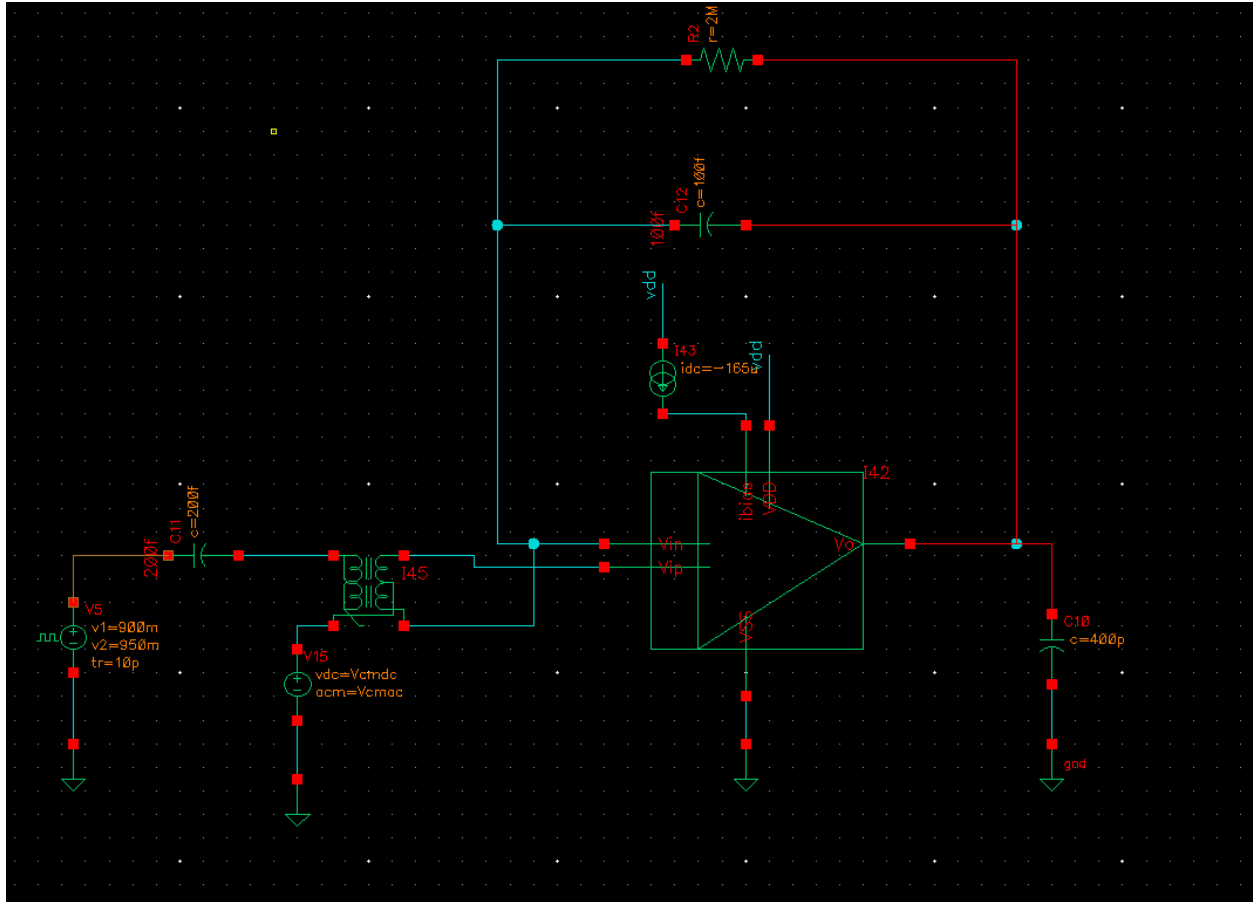
- The output matches the ideal response to within 5% of the expected step.
- If more accuracy is needed (such as for high-precision ADC or DAC applications), you'll need to reduce the op-amp's steady-state error, possibly by increasing open-loop gain or improving feedback.



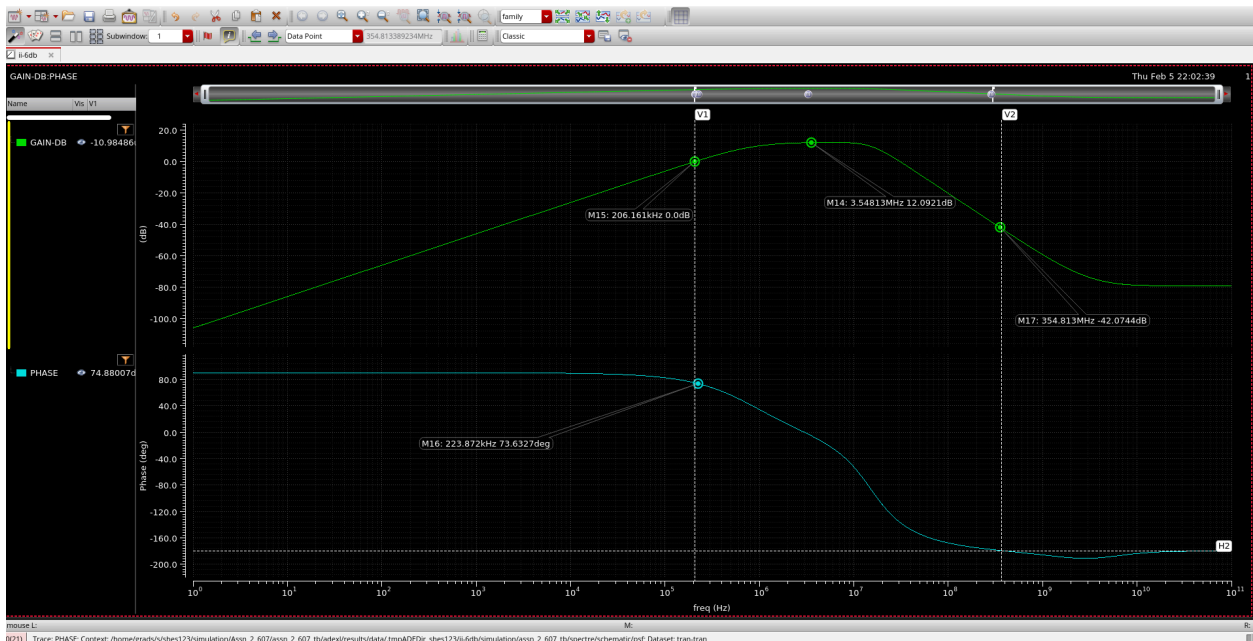
For 12db:
schematic:







Gain and Phase response:



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Session Setup Analyses Variables Outputs Simulation Results Tools Help

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Design Variables

Name	Value
1 Vdmdc	0
2 Vdmac	0
3 Vdmtr	0
4 fdm	0
5 Vcmdc	900m
6 Vcmac	0
7 p5vonly	0

Analyses

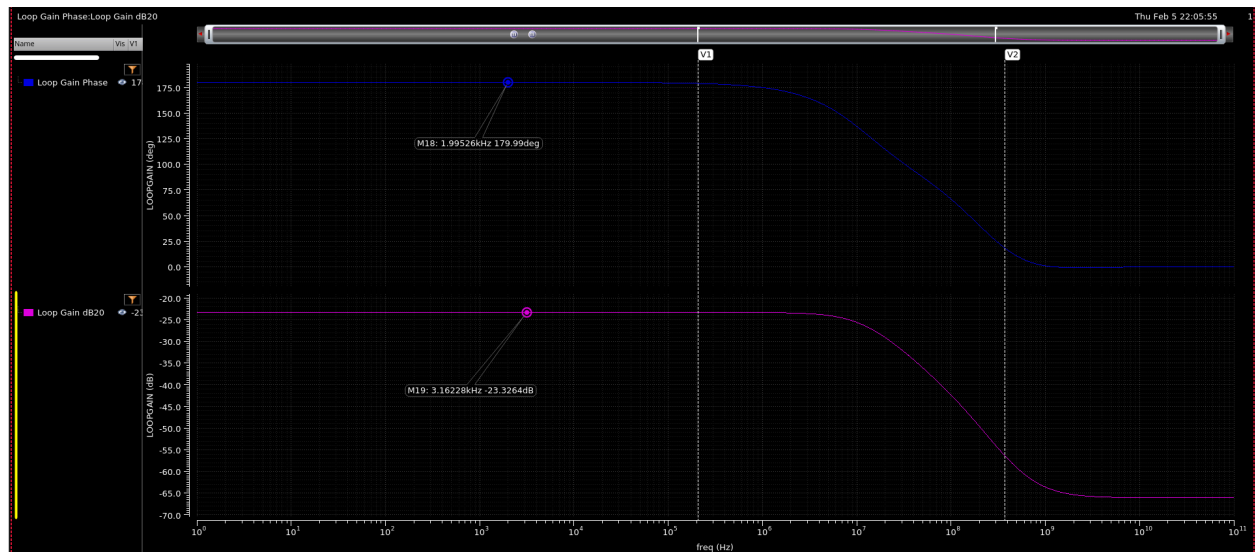
Type	Enable	Arguments
1 ac	☒	1 100G 20 Logarithmic Points Per Decade Start-Stop
2 stb	☐	1 100G 20 /IPRB2 Logarithmic Points Per Decade Sta...
3 tran	☒	0 5u

Outputs

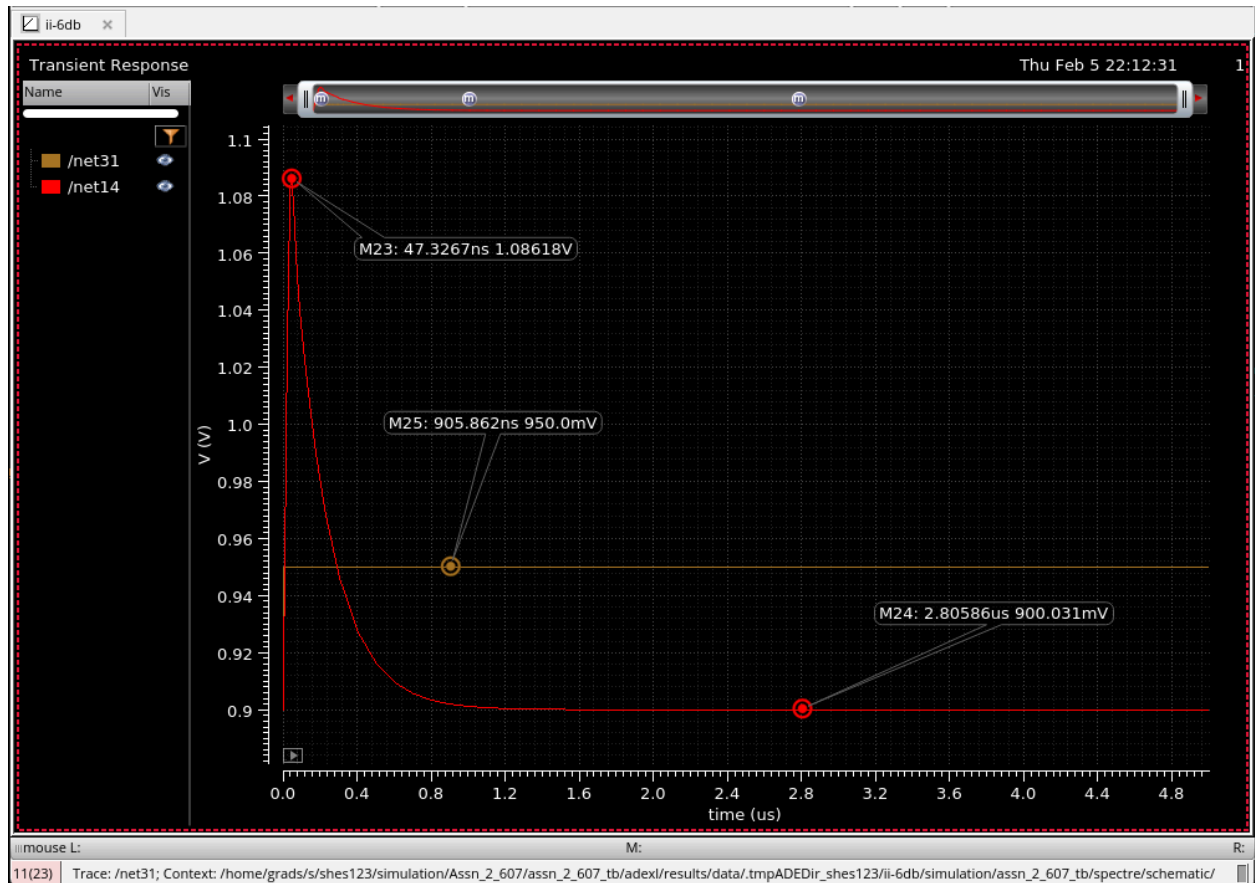
Name/Signal/Expr	Value	Plot	Save	Save Options
1 GAIN-DB		☐	☐	
2 PHASE	wave	☐	☐	
3 gain-margin	-44.43	☒	☒	
4 phase-margin	105.1	☒	☒	
5 Loop Gain Phase		☐	☐	
6 Loop Gain dB20		☐	☐	
7 net31		☒	☐	allv
8 net14		☒	☐	allv
9 net13		☐	☒	allv

> Results in ...shes123/ii-6db/simulation/assn_2_

3(4) Debug Test | Status: Ready | T=27 C | Simulator: spectre aps | State: ii-6db_active



Step response with settling time:



Input Step	0 to 50 mV
Ideal Gain (12 dB)	-3.981
Ideal Output Step	-199 mV
Initial Output	1.086 V
Ideal Final Value	0.887 V
Actual Settled Value	0.900 V
1% Band	0.885 V to 0.889 V
1% Settling Time	Not achieved
Settling Error	13 mV

For an inverting amplifier with a gain of 12 dB and an input step of 0 to 50 mV, the expected ideal output change is -199 mV (from 1.086 V to 0.887 V). The simulated step response shows

the output drops from 1.086 V and settles around 0.900 V, producing a settling (steady-state) error of 13 mV. The output does not enter the 1% settling band (0.885–0.889 V), so the 1% settling time is not achieved. This residual error is likely due to finite open-loop gain or feedback factor limitations.